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NTU ADL 2025 Fall HW2

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Instruction Tuning (Classical Chinese)

CSIE5431 Applied Deep Learning

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1 Abstract

2 LLM Tuning

2.1 Describe

1. How much training data did you use?
In the train dataset, we were using 10,000 entries for QLora Fin-Tuing in 4 bits Quantization.

2. How did you tune your model?
I basically using the Qlora method, which is from this repo: QloRa, next we are going this training file, firstly modify into my dataset abd change the parameter for improving the performance in terms of Mean perplexity by Qwen3-4B model.for the adapter setting by combining 4-bit quantization with low-rank adaptation layers, while utilizes 16-bit float to perform computations.

3. What hyper-parameters did you use?

In fig.1 we used 4 bit quantization, and next to it is the QRola parameter, how to do the Fin-Tuning.

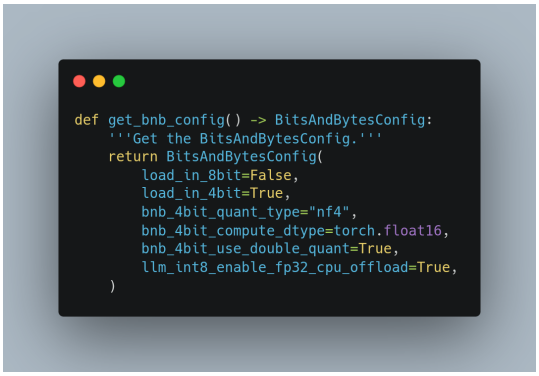


Figure 1: 4-bits



Figure 2: QLoRA

2.2 Show your performance

1. What is the final performance of your model on the public testing set?

Answer: Mean perplexity: 7.034 improved prompt given Mean perplexity: 13.16025 default prompt

2. Plot the learning curve on the public testing set:

From the fig.3, we knew the loss is already lower than 5 from steps 500, which means is already good enough for our performance.

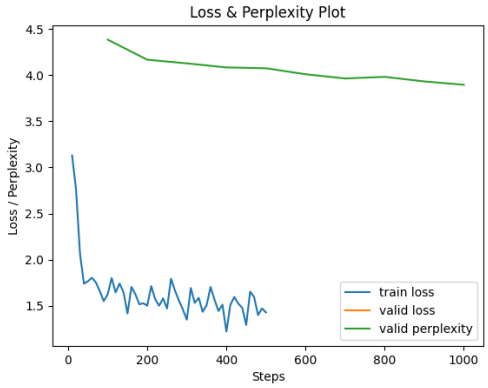


Figure 3: Learning curve

3 LLM Inference Strategies

3.1 Zero-Shot

- What is your setting? How did you design your prompt?

I evaluated Qwen/Qwen3-4B without providing demonstration examples or loading a LoRA adapter. The model was loaded with a requested bfloat16 dtype and the quantization configuration returned by get_bnb_config(). and the Evaluation used model.eval() and torh.no_grad(), so no model parameters were updated.

For each evaluation example, I restricted the input prompt using get_prompt(instruction). This provides the current instruction without example answers, allowing the experiment to measure the pretrained model's ability to follow the task instruction directly.

3.2 Few-Shot (In-context Learning)

- What is your setting? How did you design your prompt? (1%)
I used the same base model, loading configuration, maximum sequence length, and perplexity calculation as the zero-shot experiment. No model parameters were updated.

- How many in-context examples are utilized? How you select them? (1%)
For each evaluation instruction, I supplied five training examples to get_fewshot_prompt(),

together with the current instruction. The demonstrations were intended to illustrate the desired instruction response relationship and response format

3.3 Comparison

- What's the difference between the results of zero-shot, few-shot, and LoRA? (2%)

Table 1: Comparison of zero-shot, few-shot, and LoRA settings.

Method	**Model adaptation**	**Examples supplied in the prompt**
Zero-shot	No parameter updates	None
Few-shot	No parameter updates	Five training demonstrations in the corrected setup
LoRA	Trains low-rank adapter parameters while freezing the base weights	Depends on the LoRA evaluation prompt

4 Reference:

1. 4bit-transformers-bitsandbytes
2. Quantization